UNIVERSIDAD NACIONAL DE COLOMBIA

**Master's Thesis Proposal**

# High-Frequency Seismic Emission in the Quiet Sun


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Bogotá, Colombia
2025

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# 1 Motivation

There is observational evidence of high-frequency seismic emission in the Quiet Sun, and previous studies suggest that this emission originates in the solar intergranular lanes. However, this phenomenon is still not well understood, and the exact locations and physical nature of the emission sources remain uncertain.

This project aims to locate and characterize these high-frequency seismic emissions, around 11 mHz, using data from the Helioseismic and Magnetic Imager (HMI) on board the Solar Dynamics Observatory (SDO), and to compare the helioseismic results with high-resolution observations provided by the Daniel K. Inouye Solar Telescope (DKIST).

# 2 Introduction

## 2.1. Theoretical Framework

From the discovery of the 5-minute signal (see, e.g., Noyes and Leighton, 1963) and the subsequent identification of its nature as pressure waves propagating within the solar interior, the field of helioseismology was born. This discipline represents a unique opportunity to study regions of the Sun that are otherwise inaccessible to direct observation through electromagnetic radiation, due to the substantial barriers of its outer layers.

### 2.1.1. Helioseismic Holography

One of the helioseismic techniques used to probe the internal structure of the Sun is **helioseismic holography**, which was extensively developed during the 1990s (Lindsey and Braun, 1990). This technique has the major advantage of treating solar acoustic waves in an analogous way to electromagnetic optics, and unlike tomography, which suffers from poor statistics and severe diffraction effects that significantly degrade spatial resolution when applied to the solar acoustic spectrum, helioseismic holography overcomes these limitations. It has proven to be an efficient and valuable tool for locating and tracking the evolution of large-scale structures such as active regions Gonzalez Hernandez et al. (2007).

By performing the reverse holographic process, starting from the observed wave field $\psi(\mathbf{r}, t)$ at the solar surface (known as the pupil), helioseismic holography reconstructs the interior structures through which the p-modes have previously propagated (Donea et al., 2000). In this framework, a Green's function $G(|\mathbf{r} - \mathbf{r}'|, t - t')$ is introduced, which characterizes the perturbation due to the presence of an object (a source or absorber of seismic emission) and allows one to reconstruct the path that the waves have followed inside the Sun. The resulting inferred field, called the acoustic regression $H_{\pm}$, is given by:

$$H_{\pm}(\mathbf{r}, z, t) = \int dt \iint_S G(|r - r_0|, z, t \mp t')\psi(\mathbf{r'}, t')dS \tag{2-1}$$

It is possible to reconstruct the source of seismic emissions $S(r_0, t)$ by the mean egression power $\langle|\check{H}^{+}(\mathbf{r},z,\nu)|^2\rangle = \langle\check{H}^{+}(\mathbf{r},z,\nu)\check{H}^{+*}(\mathbf{r},z,\nu)\rangle$. by the integral equation:

$$\langle|\check{H}^{+}(\mathbf{r},z)|^2\rangle = \iint d^2\mathbf{r}_0\, dz_0\; S(\mathbf{r}_0,z_0)\, K(\mathbf{r},\mathbf{r}_0,z,z_0) \tag{2-2}$$

Where the kernel $K(\mathbf{r},\mathbf{r}_0,z,z_0)$ is a proper Green function representing the **egression-power signature** of a point emitter of unit acoustic luminosity at $(\mathbf{r}',z')$..
This process, referred to as the phase-coherent computational reconstruction of the acoustic field, enables the formation of subsurface images within the solar interior Donea et al. (2000).

### 2.1.2. The Helioseismic and Magnetic Imager

A key modern instrument for helioseismology is the Helioseismic and Magnetic Imager (HMI), aboard the Solar Dynamics Observatory (SDO), a solar space telescope launched in 2010. HMI records data every 45 seconds in the form of filtergrams, using six filters centered on the Fe I spectral line at 6173.3433, Å. This configuration enables precise measurements of the magnetic field strength, due to the high effective Landé factor $g_{\text{eff}}$ of the neutral iron line, as well as sensitive Doppler measurements, thanks to the narrow and relatively deep profile of this line. HMI samples both the blue and red wings of the spectral line, allowing the determination of the line-of-sight velocity (Schou et al., 2011). By comparing the filtergrams obtained at each wavelength position, Dopplergrams are constructed. Using HMI data, it is possible to apply computational seismic holography to diagnose phenomena analogous to high-frequency seismic emission in the quiet Sun.

### 2.1.3. High Frequency seismic emissions

Helioseismic observations by HMI of transient seismic emissions from solar flares suggest that seismic holography in the high-frequency p-mode spectrum can resolve the spatial distribution of seismic sources relative to the solar granulation. This is due to the fact that the solar photosphere acts as an efficient reflector for low-frequency p-modes, while high-frequency p-modes are efficiently absorbed, allowing for a more accurate determination of the source locations.

Based on this capability, computational seismic holography has been applied to high-frequency helioseismic observations of the quiet Sun from SDO/HMI in order to identify the predominant sources of seismic emission in relation to the granular structure of the photosphere Lindsey and Donea (2013). The study found that the strongest seismic emission originates at the edges of photospheric granules. Regions with continuum brightness between 95 - 100 % of the mean were about 2.5 times more emissive than those between 100 - 104 %, consistent with models in which most seismic emission arises from cool downflowing plumes at granule boundaries. Interestingly, regions with continuum brightness significantly exceeding 104 % of the mean also showed unexpectedly high seismic emissivity a feature that remains to be fully understood.

## 2.2. Statement of the Problem

Despite significant advancements in helioseismology, the mechanisms responsible for high-frequency seismic emission in the quiet Sun remain insufficiently understood. Previous studies using computational seismic holography (Lindsey and Donea, 2013) have shown that seismic emission is not uniformly distributed across the solar granulation, with enhanced emissivity observed at granular boundaries and, unexpectedly, in regions of unusually high continuum brightness. However, the physical drivers behind this dual behavior, particularly the origin of enhanced emission in overly bright regions, have yet to be clearly identified.

Furthermore, most existing analyses focus on isolated observations and do not account for potential variations across different phases of the solar cycle or the influence of small-scale magnetic structuring. As a result, it remains unclear whether high-frequency seismic emission is governed primarily by convective dynamics, magnetohydrodynamic processes, or a combination of both.

To bridge this gap, a systematic characterization of high-frequency seismic emission in the quiet Sun across different brightness regimes and magnetic environments is required. Understanding where and why seismic sources form will provide deeper insight into the coupling between granulation, magnetism, and acoustic energy generation in the solar photosphere.

# 3 Objectives

## 3.1. General Objective

To characterize the high-frequency seismic emission in the quiet Sun using computational seismic holography and to investigate its relationship with solar granulation and small-scale magnetic structuring, and to improve the reliability of helioseismic holography reconstructions and enhance detection of compact acoustic sources

## 3.2. Specific Objectives

1 1. Quantify and characterize seismic noise in SDO/HMI and high resolution DKIST observations across 3–11 mHz

2 Apply seismic holography , including filtering, spectral decomposition, spatial masking, and noise–signal discrimination in order to map the distribution of relevant seismic sources relative to continuum brightness.

3 To analyze the variation of seismic emission across different phases of the solar cycle, distinguishing between quiet and active conditions.

4 Evaluate the interplay between small-scale magnetism and seismic emission by correlating seismic-source maps with photospheric magnetograms

# 4 Methodology

This work investigates the origin of high-frequency seismic emissions in the quiet Sun by combining space-based medium-resolution observations from the *Helioseismic and Magnetic Imager* (HMI) onboard the *Solar Dynamics Observatory* (SDO) with high-resolution data from the *Daniel K. Inouye Solar Telescope* (DKIST). The methodology is organized into the description of the data sets, the preprocessing and co-alignment steps, the construction of magnetic and continuum maps, the computation of acoustic power maps using helioseismic holography, the integration of DKIST data, and the statistical analysis.

## 4.1. Data Sets and Observational Strategy

The first step of the project is to select an appropriate Quiet-Sun region. Quiet-Sun regions will be identified by masking out active regions, pores, and sunspots using DKIST images with adequate temporal coverage. In addition, only regions observed under stable and good seeing conditions will be considered.

### 4.1.1. DKIST data

High-resolution DKIST observations around 668.0 nm will be selected, ideally with time series longer than $\approx 60$ min to allow a reliable characterization of high-frequency oscillations. The chosen DKIST data set will include:

- Intensity images (and, if available, velocity and/or polarization measurements) with high spatial resolution.
- Stable seeing conditions and a consistent instrumental configuration throughout the time series.

The DKIST field of view (FOV) will be chosen so that it overlaps with the region observed by HMI, enabling a direct comparison of acoustic emission signatures at different spatial resolutions.

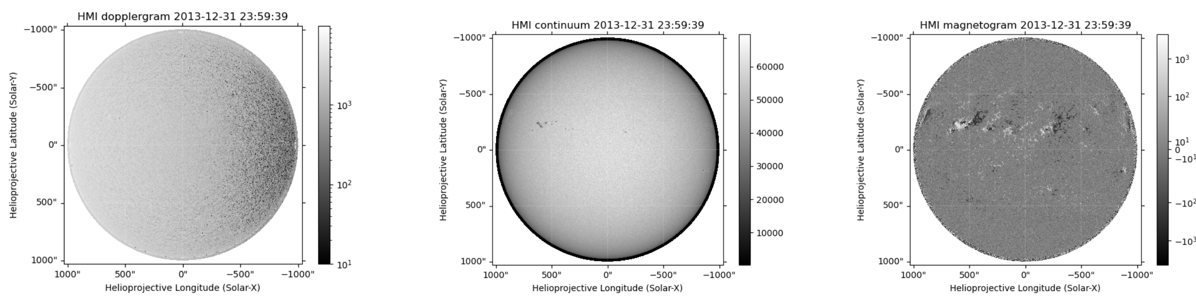
### 4.1.2. HMI/SDO data

For each selected quiet-Sun interval, the following HMI data products will be used:

- **Dopplergrams**, for helioseismic holography and the construction of acoustic power maps.

- **Continuum intensity images**, to characterize granulation and intensity fluctuations and to provide context for co-alignment with DKIST.

- **LOS magnetograms**, to characterize the underlying magnetic environment and define the quiet-Sun mask.




(a) Dopplergram (b) Continuum (c) Magnetogram

Figura **4-1**: Generated HMI maps for 2013/12/31 at 23:59:39. The dopplergram (Fig. **4-1**a), where the solar rotation can be observed, shows dark regions moving away and white regions approaching. The continuum image (Fig. **4-1**b) displays a sunspot. The magnetogram (Fig. **4-1**c) reveals the presence of strong magnetic fields and their polarity: dark for inward-directed fields and white for outward-directed fields. An active region can be seen around the sunspot showed by the continuum map, as expected.

## 4.2. Preprocessing and Co-alignment of HMI Data

### 4.2.1. Calibration and basic reduction

Standard HMI calibration procedures will be applied (flat-fielding, dark correction, distortion correction, and removal of instrumental trends) using the available pipeline products. All Dopplergram, magnetogram, and continuum intensity series will be remapped to a common coordinate system (e.g. Postel or plate carrée projection) centered on the target quiet-Sun region.

### 4.2.2. Continuum intensity maps

Co-aligned continuum intensity maps will be used to characterize the granulation pattern and its temporal evolution. Intensity contrast maps will be constructed by normalizing each continuum image to its spatial mean, thereby highlighting granules, intergranular lanes, and potential intensity anomalies. These magnetic and intensity layers will serve as reference maps to interpret the acoustic emission maps and to investigate possible correlations between high-frequency seismic sources, magnetic structures, and intensity patterns.

### 4.2.3. Magnetic maps from HMI magnetograms

For each time step, LOS magnetic field maps will be extracted and co-registered with the Doppler and continuum images. Time-averaged magnetic maps (e.g. averaged over the full time series) will be computed to characterize the mean magnetic environment of the region. Magnetic features (network vs. internetwork) will be identified by thresholding and by applying morphological operations, defining sub-regions within the quiet Sun for further analysis.

### 4.2.4. Helioseismic Holography and Acoustic Power Maps

The core of the analysis consists of computing local acoustic power maps of quiet-Sun regions applying helioseismic holography techniques to the HMI data.
A standard helioseismic holography formalism will be employed to compute the local seismic emission (egression) at the solar surface. A suitable pupil geometry (e.g. an annular pupil around the focus point) and an appropriate Green's function for acoustic waves in the solar interior will be adopted. For the desire $11mHz$ frequency band, the egression power will be computed over the quiet-Sun region, yielding maps of acoustic source power at the surface.

### 4.2.5. Integration of DKIST High-Resolution Data

#### Geometric co-alignment between DKIST and HMI

DKIST images at 668.0 nm will be geometrically co-aligned with HMI continuum intensity maps. The co-alignment will use, where available, solar limb information and/or the cross-correlation of intensity patterns (granulation) between DKIST and HMI Continuum maps. Appropriate scaling, rotation, and translation will be applied to map the DKIST field of view onto the HMI projected coordinate system.

#### Temporal matching

The DKIST time series will be synchronized with the HMI Doppler and continuum time series. Only periods where both instruments simultaneously observe the same region will be used for the joint analysis. If necessary, HMI data will be temporally interpolated to match the DKIST cadence.

#### Identification of high-frequency acoustic emission sites in DKIST

The HMI acoustic power map will be resampled onto the DKIST spatial grid to allow a point-by-point comparison. Regions of enhanced acoustic emission (above a chosen threshold, e.g. $> N\sigma$ above the mean) will be identified within the DKIST field of view. These regions will be overplotted on:

- DKIST intensity maps,
- magnetic maps (if available from DKIST or co-registered instruments),
- additional DKIST observables at 668.0 nm (e.g. line-of-sight velocity or line parameters), when available.

This comparison will allow us to examine whether high-frequency seismic sources preferentially occur in specific small-scale structures (e.g. intergranular lanes, magnetic elements, or shocks) that are unresolved by HMI.

## 4.3. Statistical Analysis

### 4.3.1. Characterization of acoustic sources

The distribution of acoustic power over the quiet-Sun region will be quantified (e.g. by histograms or probability density functions). The properties of regions with enhanced emission (size, intensity contrast, magnetic field strength, etc.) will be measured and compared across bands.

### 4.3.2. Correlation with magnetic and intensity structures

The spatial correlation between acoustic power and:

- mean magnetic field strength,
- continuum intensity,
- small-scale structures visible in DKIST,

will be quantified using, for example, Pearson or Spearman correlation coefficients, two-dimensional cross-correlation functions, and conditional averages (e.g. mean acoustic power as a function of magnetic field strength or intensity).

# 5 Chronogram

Below are the schedules for the different activities that will be carried out during the project, taking into account two academic semesters.

Cuadro **5-1**: Chronogram 2026-I

| **Activity** | **Month** | | | | | |
|---|---|---|---|---|---|---|
| | **1** | **2** | **3** | **4** | **5** | **6** |
| Bibliographic overview | | | | | | |
| Review of the computational tools | | | | | | |
| Choose good Quiet Sun samples | | | | | | |
| Produce acoustic power maps of the quiet Sun using holography | | | | | | |
| Add high-resolution DKIST data | | | | | | |
| Report findings of the first stage | | | | | | |

Cuadro **5-2**: Chronogram 2026-II

| Activity | Month | | | | | |
|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 |
| Correlate seismic data with DKIST images | | | | | | |
| Data analysis and discussion | | | | | | |
| Write the thesis document | | | | | | |

# 6 Budget

This project does not require financial support from the university. All stages of the research, including data acquisition, analysis, and documentation, will be carried out using publicly available datasets and existing institutional computational resources. Therefore, no additional economic investment is needed from the university or external funding agencies for the successful completion of this work.

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